A Statistical Analysis of Inter Twin Prime Gaps Distribution

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Introduction & Background

- **Twin Primes**: Pairs of prime numbers $(p, p+2)$ differing by exactly two, such as $(3,5)$, $(5,7)$, and $(11,13)$.
- **Twin Prime Conjecture**: Asserts the existence of infinitely many twin-prime pairs, posing an enduring open problem in analytic number theory.
- **Inter Twin Prime Gaps**: As prime numbers increase, twin primes become rarer, producing varying gap lengths between consecutive occurrences.
- **Project Scope**: Integrates computational statistics and probability theory to investigate the distributional shape, waiting-time properties, scale dependence, and extreme tails of twin-prime gaps.

Objectives of the Study

1. Generate twin-prime pairs and extract gap sequences computationally using `R`.
2. Conduct exploratory data analysis (EDA) to quantify asymmetry, variance, and heavy-tail presence.
3. Fit candidate continuous and discrete probability distributions using Maximum Likelihood Estimation (MLE).
4. Evaluate goodness-of-fit via AIC, BIC, and Kolmogorov–Smirnov (KS) statistics.
5. Investigate scale-dependent mean shift behavior across increasing prime limits.
6. Perform Extreme Value Analysis (EVA) and fit growth regression models to study tail behavior and long-run trends.

Literature Review

- **Hardy & Littlewood (1923)**: Formulated the Prime k-Tuple Conjecture, predicting that twin prime density decreases asymptotically as $1/(\log x)^2$.
- **Yitang Zhang (2014) & Maynard**: Proved the existence of bounded gaps between primes, marking major progress in prime spacing theory.
- **Cohen (2024)**: Analyzed prime gaps using exponential approximations, demonstrating strong suitability for waiting-time data.
- **Afriyie (2025)**: Applied Extreme Value Theory (EVT) to twin prime gaps, showing heavy-tailed properties in upper quantiles.

Methodology Framework

Twin Prime Gap Definition

For consecutive twin-prime pairs $(p_i, p_i + 2)$ and $(p_{i+1}, p_{i+1} + 2)$, the gap G_i is defined as:

$$G_i = p_{i+1} - (p_i + 2)$$

- **Computational Setup**: Implemented in `R` (v4.5.0) using the `primes` package.
- **Analytical Pipeline**:
  - Exploratory visualization (Histograms, ECDF, Boxplots, Log-Frequency plots).
  - Parametric distribution fitting via MLE.
  - Scale comparison, OLS growth regression, and Peaks-Over-Threshold (POT) EVA.

Dataset Description ($N = 10^8$)

- Primary statistical analysis conducted on prime pairs up to upper bound $N = 10^8$.
- Provides a sufficiently large sample size for robust tail estimation and parametric inference.

Table 1: First 7 Twin Prime Pairs and Gap Calculations

Index (i)	**Twin Prime Pair**	**Prime (p_i)**	**Gap (G_i)**
1	$(3, 5)$	3	–
2	$(5, 7)$	5	0
3	$(11, 13)$	11	4
4	$(17, 19)$	17	4
5	$(29, 31)$	29	10
6	$(41, 43)$	41	10
7	$(59, 61)$	59	16

Exploratory Analysis: Descriptive Statistics

Sample Metrics ($N = 10^8$)

- **Sample Mean ($\bar{G}$)**: 225.1
- **Variance (S^2)**: 47018.18
- **Skewness**: 1.9937
- **Kurtosis**: 9.0394

Key Statistical Insights

- $S^2 \gg \bar{G}$: Demonstrates severe **over-dispersion** in gap length.
- Positive skewness (1.99) establishes an asymmetric distribution skewed right.
- High kurtosis (9.04) indicates a leptokurtic shape with heavy tail occurrences.

EDA: Histogram and Kernel Density Analysis

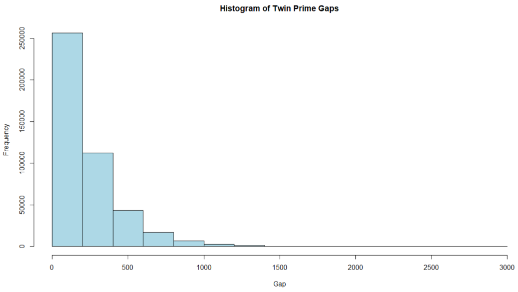

Figure 1: Histogram and Kernel Density Estimate of Twin Prime Gaps

Remarks from Figure

- Displays a unimodal, strongly right-skewed distribution.
- Frequencies concentrate heavily at lower gap values and decline rapidly as gap size increases.
- Extended upper tail indicates waiting-time characteristics typical of exponential-type families.

EDA: ECDF and Boxplot Diagnostics

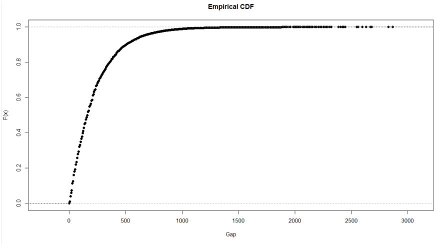

Figure 2: Empirical CDF

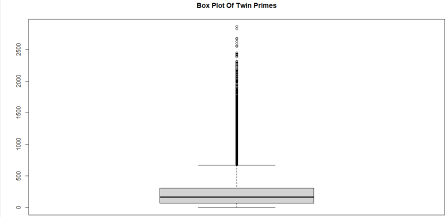

Figure 3: Boxplot of Gaps

Remarks from Figures

- **ECDF**: Rises steeply for small gaps and approaches 1.0 at a diminishing rate.
- **Boxplot**: Highlights numerous upper-tail outliers. These reflect genuine arithmetic properties of prime distributions rather than erroneous observations.

EDA: Log-Frequency Diagnostic Analysis

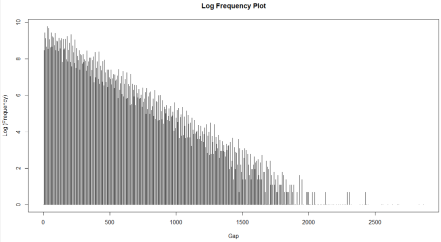

Figure 4: Log-Frequency Plot of Twin Prime Gaps

Remarks from Figure

- Under exponential decay $f(g) = Ce^{-\lambda g}$, the log-frequency relation $\log f(g) = \log C - \lambda g$ should be linear.
- The plot exhibits approximate linearity with negative slope $-\lambda$, confirming waiting-time suitability.
- Minor tail departures motivate flexible models like Gamma or Weibull.

Candidate Continuous Probability Distributions

Four continuous models evaluated via Maximum Likelihood Estimation (MLE):

- **Exponential Distribution**:

$$f(x;\lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

- **Gamma Distribution**:

$$f(x;\alpha,\beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

- **Weibull Distribution**:

$$f(x;k,\lambda) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, \quad x > 0$$

- **Log-normal Distribution**:

$$f(x;\mu,\sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log x - \mu)^2}{2\sigma^2}\right], \quad x > 0$$

Methodological Detail: Exclusion of Zero-Gap Frequency

- Up to $N = 10^8$, exactly **one** zero gap occurs ($G_1 = 0$, between twin pairs $(3, 5)$ and $(5, 7)$).
- Continuous waiting-time models face mathematical boundary issues at $x = 0$ (e.g., $\log(0)$ is undefined for Log-normal).
- Forcing $x = 0$ distorts global likelihood optimization ($\ln \hat{L}$).
- **Statistical Outcome**: Excluding $x = 0$ significantly lowers AIC and BIC across continuous specifications, enabling accurate representation of asymptotic behavior.

Continuous Distribution Goodness-of-Fit Comparison

Table 2: Model Evaluation Statistics (Excluding Zero Gap)

Distribution	**AIC**	**BIC**	**KS Statistic**	**KS *p*-value**
Exponential	5,650,586	5,650,597	0.04378	$< 10^{-16}$
Gamma	**5,645,470**	**5,645,492**	0.02666	2.62×10^{-272}
Weibull	5,646,953	5,646,975	**0.02517**	1.21×10^{-242}
Log-normal	5,681,908	5,681,930	0.07105	$< 10^{-16}$

Model Selection Strategy

While Weibull produces the lowest KS statistic, the **Gamma distribution attains the lowest AIC and BIC**, providing the optimal balance of goodness-of-fit and model parsimony.

Graphical Diagnostics: Q-Q and Overlay Plots

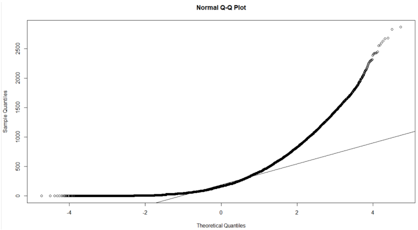

Figure 5: Normal Q-Q Plot

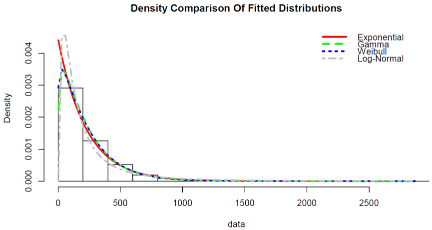

Figure 6: Density Comparison

Remarks from Figures

- **Normal Q-Q Plot**: Severe departures in upper quantiles confirm non-normality.
- **Density Comparison**: Fitted Gamma density aligns most closely with empirical data across central and tail ranges.

Discrete Probability Models vs Continuous Selection

Table 3: Goodness-of-Fit for Discrete Models ($N = 10^8$)

Distribution	Parameter Estimate	Log-Likelihood	AIC	BIC
Poisson	$\hat{\lambda} = 225.1113$	$-40,281,139$	$80,562,281$	$80,562,292$
Geometric	$\hat{p} = 0.004422$	$-2,826,274$	$5,652,550$	$5,652,561$

- **Poisson Model**: Fails due to severe over-dispersion ($\mathrm{Var}(X) \gg E(X)$).
- **Geometric Model**: Better than Poisson, but assigns non-zero probabilities to odd gaps (which are impossible for prime gaps > 2).
- **Final Model Choice**: **Gamma distribution** is selected as the primary probabilistic model.

Mean Shift Analysis Across Scale Bounds

Table 4: Estimated Gamma Parameters Across Prime Limits

Prime Limit (N)	**Shape (α)**	**Rate (β)**	**Mean Gap (α/β)**
10^4	1.379092	0.028941	47.65
10^5	1.348944	0.016703	80.76
10^6	1.267051	0.010432	121.46
10^7	1.214988	0.007208	168.56
10^8	1.148650	0.005101	225.17

Empirical Finding

The estimated mean gap expands continuously with larger prime limits, demonstrating scale non-stationarity.

Theoretical Justification of Mean Shift

- Hardy–Littlewood Twin Prime Conjecture states local density $\lambda(x)$ near x is:

$$\lambda(x) \approx \frac{2C_2}{(\log x)^2}, \quad C_2 \approx 0.66016$$

- Expected spacing D_x between consecutive twin prime pairs scales as:

$$E(D_x) \approx \frac{1}{\lambda(x)} = \frac{(\log x)^2}{2C_2}$$

- Accounting for twin pair width gives expected gap $E(G_x)$:

$$E(G_x) \approx \frac{(\log x)^2}{2C_2} - 2$$

- Proves theoretically that average gap size must increase logarithmically as $x \to \infty$.

Visualizing Distributional Drift (Mean Shift)

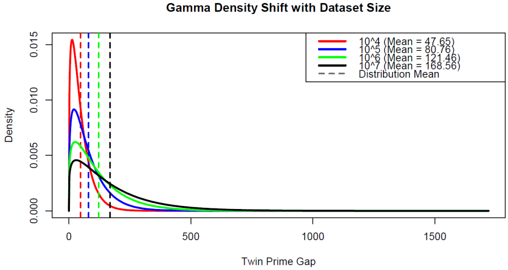

Figure 7: Gamma Density Overlay Across Prime Limits (10^4 to 10^8)

Remarks from Figure

- Fitted Gamma densities shift systematically rightward as N increases.
- Peak density flattens while scale and mean parameters expand, corroborating the Hardy–Littlewood theoretical prediction.

Regression Analysis: Growth Model Fitting

Table 5: Comparison of Growth Regression Models

Model Specification	**Regression Equation**	R^2	**AIC**	**BIC**
Gap vs Index (I)	$\widehat{G} = 86.95 + 0.008195I$	0.0289	100264.5	100285.5
Gap vs Prime (p)	$\widehat{G} = 91.77 + 0.00006306p$	0.0272	100278.8	100299.9
Gap vs $\log(p)$	$\widehat{G} = -92.33 + 16.89\log(p)$	0.0349	100213.7	100234.7
Gap vs $(\log p)^2$	$\widehat{G} = -0.568 + 0.755(\log p)^2$	**0.0355**	**100208.5**	**100229.5**

- Positive regression slopes are statistically significant ($p < 2.2 \times 10^{-16}$) across all models.
- The $(\log p)^2$ specification produces the highest R^2 and lowest AIC/BIC.

Regression Diagnostics: Gap vs $(\log p)^2$

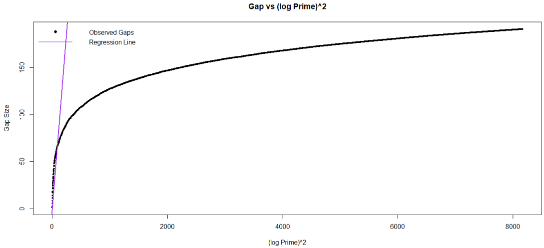

Figure 8: Twin Prime Gap vs $(\log p)^2$ with Fitted Line

Remarks from Figure

- High individual scatter around the regression line yields a modest $R^2 = 3.55\%$.
- The positive slope reflects long-run average growth rather than predicting individual gap lengths.

Extreme Value Analysis: Threshold Selection

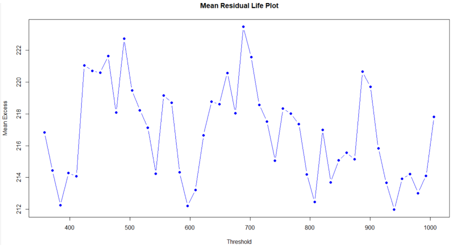

Figure 9: Mean Residual Life (MRL) Plot

Remarks from Figure

- MRL function $e(u) = E(X - u \mid X > u)$ exhibits stability across threshold range 450–750.
- Supports Generalized Pareto Distribution (GPD) modeling above threshold $u = 658$ (95th percentile).

Generalized Pareto Distribution (GPD) Fitting

Exceedances $Y = X - u$ above threshold u modeled using GPD:

$$F(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0$$

Table 6: GPD Parameter Estimates Across Percentile Thresholds

Percentile	Threshold (u)	Shape (ξ)	Scale (β)	Exceedances
95%	**658**	**231.41**	**−0.02995**	**21,171**
97%	766	223.34	−0.01513	13,198
99%	1006	217.87	−0.00008	4,375

Decision

The **95th percentile ($u = 658$)** preserves adequate sample size ($21,171$ exceedances) while ensuring parameter stability.

GPD Diagnostics: Tail Model Validation

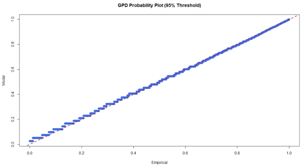

Figure 10: Probability Plot

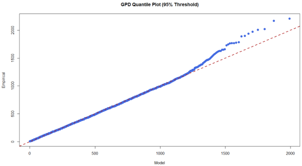

Figure 11: Quantile Plot

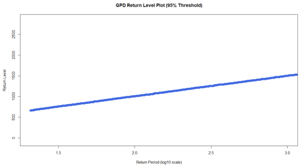

Figure 12: Return Level

Remarks from Figures

- Empirical points adhere closely to the linear reference lines in Probability and Q-Q plots.
- Return level curve captures extreme upper tail behavior within confidence bounds.

Conclusions, Limitations & Future Scope

Key Conclusions

- Twin-prime gaps exhibit positively skewed waiting-time properties best captured by a **Gamma distribution**.
- Average gap growth scales as $(\log p)^2$, supporting Hardy–Littlewood theoretical predictions.
- Extreme gaps strictly follow Generalized Pareto behavior under POT.

Limitations:

- Computational upper bound limited to $N = 10^8$.
- Statistical modeling provides empirical evidence, not formal proof.

Future Scope:

- Extend to Cousin $(p, p+4)$ and Sexy $(p, p+6)$ prime pairs.
- Apply Bayesian inference and survival modeling.

References & GitHub Repository

Key References

- Cohen, J. E. (2024). *Gaps Between Consecutive Primes and the Exponential Distribution*. Experimental Mathematics.
- Zhang, Y. (2014). *Bounded Gaps Between Primes*. Annals of Mathematics.
- Afriyie, G. (2025). *Extreme Value Theory Applied to Twin Primes...*. SSRN.

GitHub Repository

- **Link**: `https://github.com/sayan05on-ctrl/Twin-Prime-Gap-Analysis-And-Extreme-Value-Modeling`
- Contains full R source code, datasets, and execution scripts.